Attention-Based Mean-Max Balance Assignment for Oriented Object Detection in Optical Remote Sensing Images

Qifeng Lin, Nuo Chen, Haibin Huang, Daoye Zhu, Gang Fu, Chuanxi Chen, and Yuanlong Yu

***Abstract*— For objects with arbitrary angles in optical remote sensing (RS) images, the oriented bounding box regression task often faces the problem of ambiguous boundaries between positive and negative samples. The statistical analysis of existing label assignment strategies reveals that anchors with low Intersection over Union (IoU) between ground truth (GT) may also accurately surround the GT after decoding. Therefore, this article proposes an attention-based mean-max balance assignment (AMMBA) strategy, which consists of two parts: mean-max balance assignment (MMBA) strategy and balance feature pyramid with attention (BFPA). MMBA employs the mean-max assignment (MMA) and balance assignment (BA) to dynamically calculate a positive threshold and adaptively match better positive samples for each GT for training. Meanwhile, to meet the need of MMBA for more accurate feature maps, we construct a BFPA module that integrates spatial and scale attention mechanisms to promote global information propagation. Combined with S2ANet, our AMMBA method can effectively achieve state-of-the-art performance, with a precision of 80.91% on the DOTA dataset in a simple plug-and-play fashion. Extensive experiments on three challenging optical RS image datasets (DOTA-v1.0, HRSC, and DIOR-R) further demonstrate the balance between precision and speed in single-stage object detectors. Our AMMBA has enough potential to assist all existing RS models in a simple way to achieve better detection performance. The code is available at https://github.com/promisekoloer/AMMBA.**

***Index Terms*— Attention feature fusion, label assignment, optical remote sensing (RS) images, oriented object detection.**

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I. Introduction

THE detection of arbitrarily oriented objects represents a significant advancement over traditional horizontal object detection in computer vision, especially in applications such as ground target recognition in remote sensing (RS) images, which require high positional and boundary accuracy. Oriented object detection in RS includes classification and regression tasks, using oriented bounding boxes (OBBs) instead of horizontal bounding boxes (HBBs) to accurately locate objects. OBBs encode five-tuple information (x, y, w, h, a), representing the center coordinates, width, height, and angle to capture object orientation and size effectively. Many horizontal object detectors have transitioned to oriented detection tasks [1], [2], [3], [4], [5], [6].

In regression tasks, positive samples are anchor boxes with higher overlap to the GT, used to refine the target's boundaries and position. Negative samples, with little overlap, are used in classification to indicate the absence of targets. The lack of clear boundaries between positive and negative samples, along with arbitrary target directions and scale variations, complicates label assignment, making it crucial for model training.

Currently, the Max-IoU assignment strategy (MaxIoU for simplicity), based on the Intersection-over-Union (IoU) method, is widely used in RS detectors [3], [7], [8]. It assigns anchors as positive samples for GTs when the IoUs of these anchors exceed a certain threshold or, otherwise, only the anchor with the highest IoU is assigned positive. However, the MaxIoU strategy is not suitable for oriented object detection in RS images for the following reasons.

1) GT boxes have arbitrary directions, resulting in lower IoU with horizontal anchor boxes and causing misalignment issues.
2) If a GT lacks positive anchors above the fixed IoU threshold, relying on a single lower IoU positive sample may lead to inadequate training.
3) The IoU threshold significantly impacts model performance, often requiring redundant experiments to determine the appropriate threshold for datasets with diverse distributions.

The quality of label assignment in regression tasks is influenced by several factors, including selecting an appropriate IoU threshold [9], setting different positive label thresholds for GTs with varying characteristics [10], and accounting

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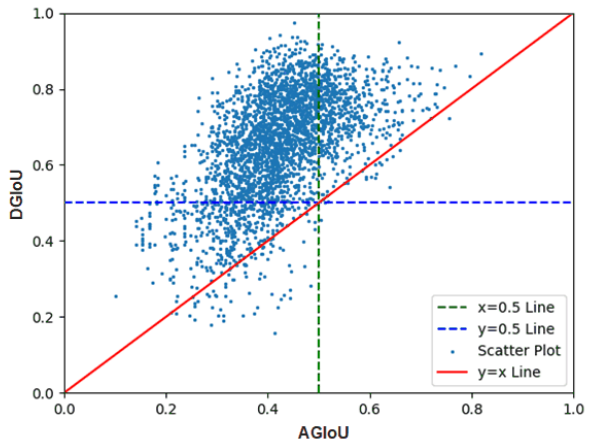


Fig. 1. Scatter plot of AGIoU (IoU between anchor and GT) and DGIoU (IoU between the decoded box and GT). We loaded the weights of the 12th epoch of RetinaNet trained on the DOTA dataset using the MaxIoUAssign strategy and setting the positive IoU threshold to 0.5.

for the varying quality of positive samples by adjusting their gradient propagation weights [2]. To address the limitations of MaxIoU-based approaches, several studies have proposed improvements from different perspectives. Some works [11], [12] focus on enabling the model to first learn how to generate suitable anchors. Other strategies [10], [13], [14], [15] employ probabilistic or statistical methods to match anchors with GTs, maximizing either global or local probabilities. In addition, methods such as [16] and [17] assign weights to samples of varying quality during loss calculation, thereby stabilizing the model's training process.

However, we observe that the misalignment between anchors and GTs is not a necessary problem that requires addressing by placing a large number of anchors with different sizes and ratios. This comes from a pre-experiment conducted by loading the weights of a well-trained RetinaNet [3], which adopts the MaxIoU strategy with a positive sample threshold set to 0.5 for training. For simplicity, we refer to the IoU between the anchor and GT as AGIoU and the IoU between the decoded box and GT as DGIoU. When only generating one anchor per feature position, we observe that even preset anchors with lower AGIoU values retain the potential for higher DGIoU. Fig. 1 illustrates that among accurately predicted decoded bounding boxes (DGIoU $>$ 0.5), 78.23% were generated by poorly aligned anchors (AGIoU $<$ 0.5). Moreover, within this subset of poorly aligned anchors, 81.49% corresponds to decoded bounding boxes that are accurately predicted. These indicate that a considerable portion of anchors with good potential for regression is excluded by the fixed positive IoU threshold of 0.5. As depicted in Fig. 2, three anchors demonstrating strong localization and regression potential are categorized as negative samples because they fail to meet the positive IoU threshold. For simplicity, we use A_1, A_2, and A_3 to represent the red, blue, and orange anchors, respectively. Despite having the best-matched box with overlapping center points, A_1 shows a lower IoU value due to the angle mismatch of the GT. While A_2 and A_3 have the same IoU value, their learnable ranges of features for the GT differ, as reflected in the varying distances between the center points of anchors and the GT. Disregarding potential positive anchors with lower IoU values but more accurate positioning or bounding hampers the model's ability to learn effectively from positive samples, particularly for objects with arbitrary angles in RS images.

To solve the problem of insufficient learning of positive samples, this article proposes an attention-based mean-max balance assignment (AMMBA) strategy, consisting of a mean-max balance assignment (MMBA) strategy and a balance feature pyramid with attention (BFPA) module. MMBA establishes sets of candidate anchors for each GT and uses the mean-max assignment (MMA) to determine a dynamic positive IoU threshold. A balance assignment (BA) strategy leverages the centerness branch to generate weights for balancing the loss calculation between high- and low-quality positive samples. Meanwhile, considering the fact that MMBA may reduce the number of generated anchors, higher requirements are placed on feature map quality. To enhance the localization learning of center features, we design the BFPA module that uses attention mechanisms to enhance spatial and shape information, increase information flow between adjacent layers, and integrate features from each layer to improve the global semantic information of the feature map.

The main contributions of this work can be summarized as follows.

1) A significant finding from our analysis is that the anchors with lower IoU between ground truths (GTs) still have great potential to precisely enclose the GTs after decoding.
2) A novel MMBA is proposed to assign more and better positive samples for sufficient learning of the detector.
3) A BFPA module is developed to significantly enhance the model's feature learning capabilities for the MMBA strategy.
4) The proposed AMMBA is a plug-and-play design, which can be seamlessly integrated into anchor-based detectors, thereby achieving state-of-the-art results on challenging RS image datasets.

II. Related Works

A. Label Assignment Strategy of Regression Samples

Current assignment strategies generally fall into two categories: hard-label assignment and soft-label assignment.

1) Hard Label Assignment: The anchor-based detector, the most common and effective hard label assignment method, is MaxIoU. It is obvious that this strategy is intuitive and simple and is widely used in label allocation strategies for rotating object detectors [3], [7], [18], [19], [20], [21]. Certainly, the IoU threshold hyperparameter will be designed with different values for different stages or detectors.

Some anchor-free detectors utilize the distance from the center points as a measurement such as FCOS [2] and Foveabox [22], while others [23] directly calculate the chamfer distance between the set closures of key points as the boundary between positive and negative samples. As our focus is on the anchor-based label assignment strategy, we will not discuss anchor-free methods in detail.

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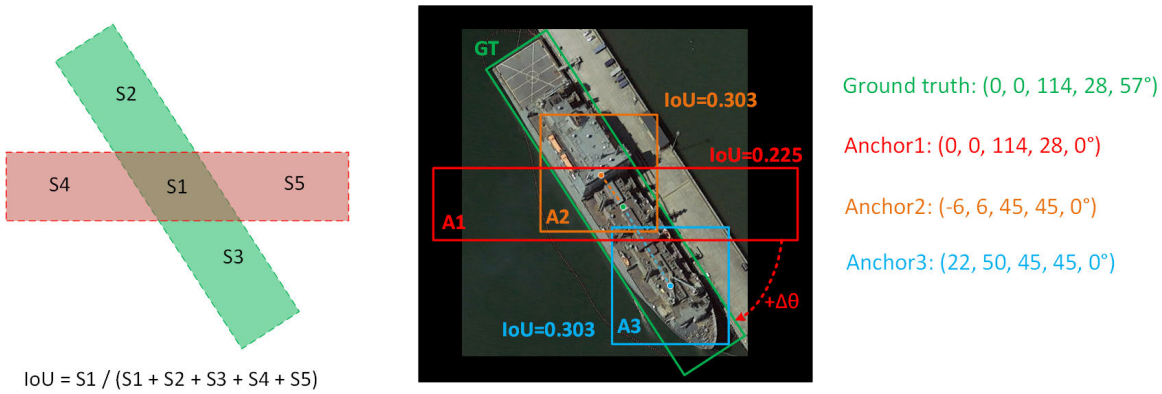


Fig. 2. These three preset anchors all have good potential for localization and regression, but they are all assigned as negative samples under the IoU > 0.5.

The hard assignment method may hinder the learning of positive samples since there is no theoretical evidence to suggest that setting hyperparameters to specific values is optimal. Therefore, these parameters can only be determined through extensive experimentation and may depend on the distribution of shapes of the identified objects, rendering them unsuitable for applying the same hyperparameter values across all datasets.

2) Dynamic Label Assignment: In the soft label assignment strategy, the primary concept is to pair anchor boxes with GTs based on probability and statistical algorithms.

These approaches involve designing different positive thresholds for various GTs. OTA [14] proposes to shift from independently pursuing optimal allocation for each GT to global optimization. ATSS [10] addresses the issue of imbalanced samples by automatically calculating distinct threshold limits for different targets based on the statistical characteristics of objects. SASM [15] utilizes the aspect ratio of the GT as a control parameter to dynamically calculate positive and negative samples based on ATSS. FreeAnchor [24] utilizes the process of maximum likelihood estimation to select the most representative anchor box from a set of candidate anchor boxes as its positive sample. GGHL [25] proposes a target adaptive sampling strategy (OLA) based on a 2-D rotating Gaussian heatmap, which more accurately reflects the target's size and directional characteristics.

While others assign different weights to samples based on their quality in the loss function, DAL [17] combines the IoU between the anchor box and the GT before regression with the IoU of the bounding box and the GT after regression to set compensation values for learning samples with good localization but low actual IoU. Focal loss [3] reduces the weight of easy-to-classify targets and increases the weight of difficult-to-classify targets, focusing more on difficult-to-classify targets and enhancing the robustness of the model. GFL [26] combines the quality of classification and localization to match soft weights for each positive anchor box. DW [16] strategy not only designs positive weights for samples but also takes the probability and importance of samples as two negative weight indicators.

However, they did not consider the difficulty of learning samples of the same category but with different sizes, and the shape of the target can also affect the label assignment process. In this work, we draw inspiration from the statistical method of ATSS [10], SASM [15], and the mean-max function in FreeAnchor [24]. Then, we propose the MMBA strategy, which accounts for the difficulty of identifying a GT through the anchors assigned to it, recalculates the positive IoU threshold, and uses the center score to weight the loss calculation for different quality positives.

B. Feature Compensation Module

Common backbones for object detection tasks include VGG [27], ResNet [28], DenseNet [29], and ResNeXt [30], as well as more recent architectures based on the Transformer [31]. These backbones extract features by utilizing either convolutional blocks or self-attention mechanisms in multilayer structures to capture semantic information. Traditional CNN-based methods have been widely adopted in object detection [1] due to their efficiency in processing spatial hierarchies and multiscale features. However, one challenge faced by both CNN and Transformer-based architectures is the tradeoff between rich semantic information with low resolution in deep feature layers and high resolution with insufficient semantic information in shallow feature layers, which hinders improvements in detector accuracy.

To address these challenges, FPN [32] proposes horizontal feature compensation, enriching shallow semantic information through top-down bypasses. The PAFPN [33] enhances the entire feature pyramid with precise positioning signals from lower layers, shortening the information path from bottom to top. NPMMRDet [34] directs a neural network model to prioritize relevant features and minimize background noise. Libra R-CNN [35] introduces a balanced feature pyramid (BFP) module, which reshapes each layer to the same size and fuses them to enhance global information. Dynamic head [36] refers to the design of a DyHead block, which undergoes convolution using size-, spatial-, and task-aware attention mechanisms to concentrate the feature map more on the targets. In addition, several works [37], [38], [39] have demonstrated the effectiveness of pyramid structures in multiscale training and have applied them as foundational structures in the training process.

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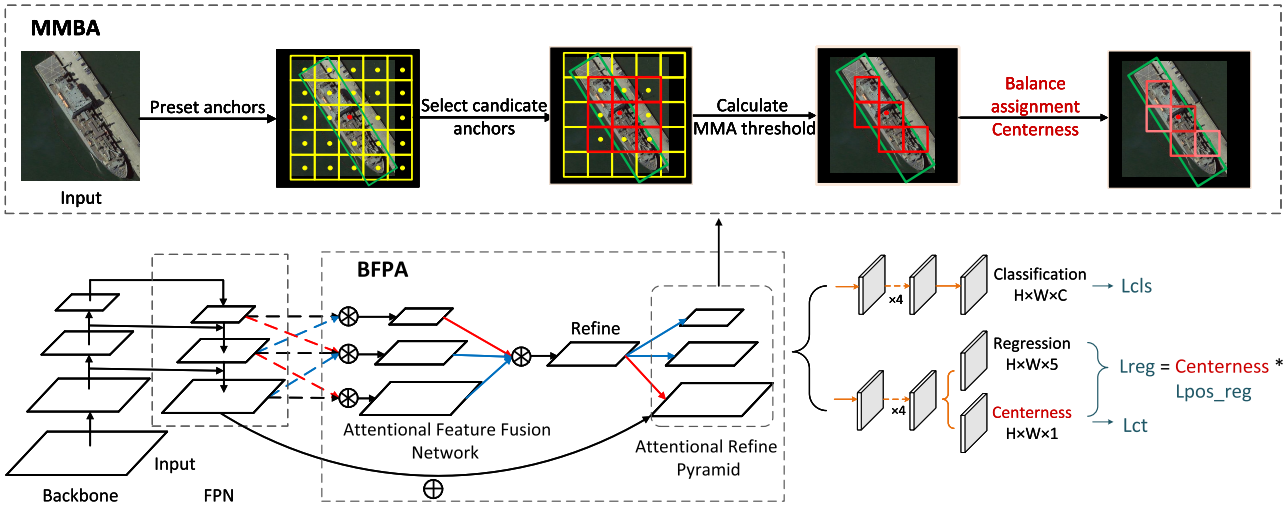


Fig. 3. Overall architecture of the proposed detector consists of a backbone, a feature pyramid network (FPN), an AFFN, and an attentional refine pyramid (AFP). For simplicity, only three feature layers are depicted although five layers are actually constructed.

In this article, we propose the BFPA module, which effectively integrates spatial and shape attention mechanisms, along with feature fusion, to enhance the flow of information between adjacent layers. The attention mechanism improves spatial and shape recognition, which is essential for effectively learning low-matching anchors designated as positive samples in the MMBA strategy.

III. Method

A. Overall Architecture

The detection architecture is primarily based on the RetinaNet [3] structure, as shown in Fig. 3. The proposed MMBA employs the MMA strategy to allocate prior anchors to positive and negative regression samples. Furthermore, inspired by FCOS [2], the BA in MMBA is used for weights balancing the assigned label learning. Specifically, centerness scores from BA serve as weights for balancing the loss calculation between high- and low-quality positive samples. Positive samples with higher centerness scores are assigned larger weights, which helps suppress the gradient of samples with lower matched samples. To further enhance the learning of center localization features within the MMBA strategy, we introduce the BFPA module. It consists of two parts: the attentional fusion feature network (AFFN) and the attentional refinement pyramid (ARP). Specific implementation details will be elucidated in Sections III-B and III-C.

B. MMBA Strategy

The mean-max function [24] exhibits a distinct characteristic: as the maximum value in the set approaches the predefined maximum value (set to 1 in this work, representing the maximum IoU value), the function output tends to approach the maximum value in this set. Specifically, for simplicity, we chose just two inputs, x and y, where x is a fixed value in [0, 0.2, 0.4, 0.6, 0.8, 1] and y varies continuously between 0 and 1, as shown in Fig. 4. The red dashed line indicates

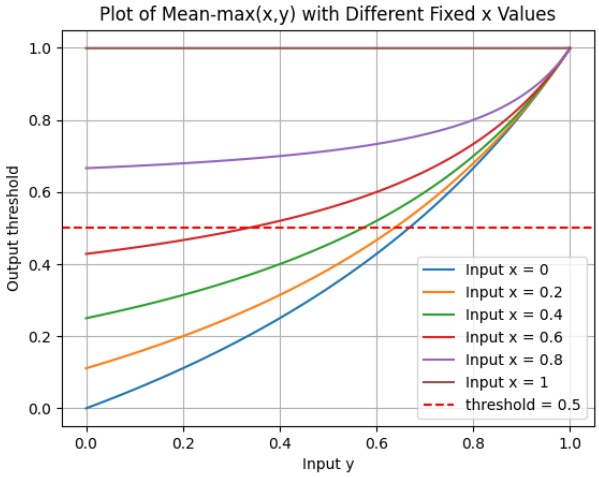


Fig. 4. Graph of a mean-max function, with a fixed input x and a variable y, illustrates that when one or more inputs exceed the threshold of 0.5, the output approaches the maximum of the group. While the group's variance is lower and all inputs are below the threshold, the output is closer to the average. In our experiments, the inputs consist of AGIoUs.

that the IoU threshold is 0.5, which is considered a positive sample threshold in MaxIoU. It can be observed that as the fixed input x value increases, the value of y only requires a smaller value to obtain a larger output value. Similarly, as the y value increases, the output value suddenly rises. However, the range of output values is limited between the values of x and y. In practice, however, the mean-max function has multiple inputs, specifically the AGIoUs between the GT and the closest anchors, as discussed later.

We propose the MMBA strategy, which enhances the MMA strategy by incorporating BA. Thus, we introduce the MMA strategy in detail first. MMA predefines a fixed-size square anchor box at each position on each layer of the feature map. Then, we use the L2 distance metric to calculate the distance from the center of the GT box to each anchor, selecting the

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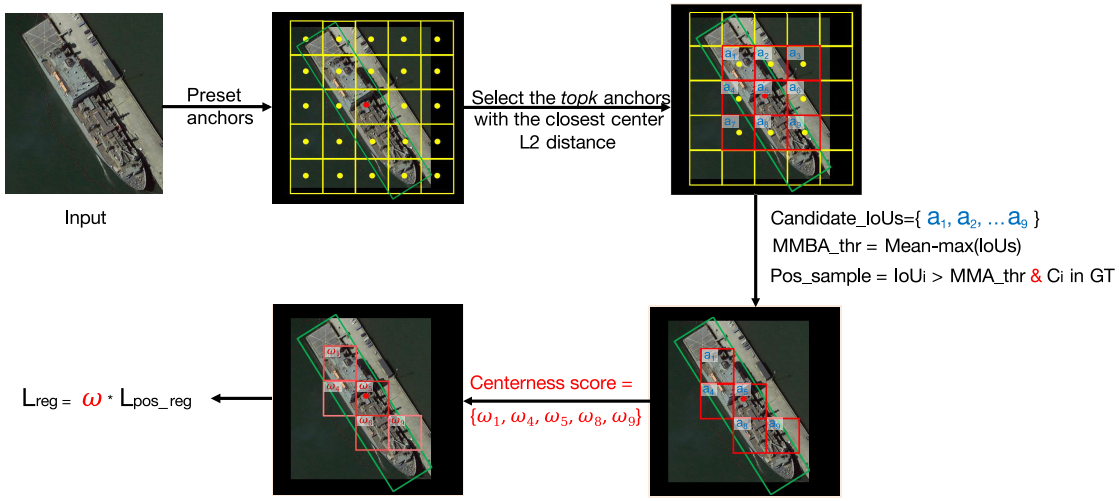


Fig. 5. Total pipeline of the MMBA strategy.

top k (in short: *topk*) nearest anchors at each feature level. Supposing that there are Y feature pyramid levels, for each GT box, a set of candidate anchors will be formed with top$k$ $\times$ Y candidate anchors. The hyperparameter *topk* is set to 9 by default. However, unlike the methods mentioned above [10], [15], which rely on the mean and variance of the candidate anchors' IoU to generate a new IoU threshold for separating positive and negative candidate anchor boxes for each GT box, MMA employs the mean-max function for this purpose. Our mean-max threshold (MM-Thr) calculation formula is given as follows:

$$\text{MM-Thr}(\text{GT}_i) = \text{Mean-max}(\text{IoU}_{i,j}) = \frac{\sum_{x_i \in \text{IoU}_{i,j}} \frac{x_i}{1-x_i}}{\sum_{x_i \in \text{IoU}_{i,j}} \frac{1}{1-x_i}} \quad (1)$$

where $\text{IoU}_{i,j}$ represents the set of the IoU between the candidate anchors and GT_j. To address potential outliers, we restrict the MM-thr range to between 0.005 and 0.5. Finally, only candidate anchor$_i$ with $\text{IoU}_{i,j} >$ MM-Thr(GT_j) and a center point within GT_j is assigned as a positive sample. Algorithm 1 describes the MMA procedure for assigning positive and negative samples in detail.

Based on MMA, the more effective positive samples are obtained for learning. However, we find a fact that different positive samples have different effects on the semantic learning of the detector. Introducing positive anchors far away from the center of GT will cause unimportant gradient interference, resulting in a decrease in detection accuracy after training. Therefore, how to better leverage these positive samples becomes another key to improving model training. Therefore, a BA strategy is proposed to weigh whether the anchors are far away from the center of the object, so as to achieve balanced gradient propagation between different positive samples. Our BA leverages the centralization branch of 1 $\times$ 1 convolution to distinguish the quality of the positive samples' regression. This BA strategy incorporates weights into the calculation of regression loss and assigns different weights to positive samples according to the quality of positive samples indicated by the centralization score. By suppressing the gradient learning of low-quality positive samples, the whole MMBA (including MMA and BA) achieves balanced distribution and stable training. The MMBA strategy process is shown in Fig. 5.

Algorithm 1 MMA Procedure

GT is a set of ground-truth boxes on the image
L is the overall levels of feature pyramid
A_i is a set of candidate anchors from the ith pyramid level
A is a set of all levels candidate anchors
$Topk$ is a hyperparameter with a default value of 9
P is a set of positive samples
N is a set of negative samples
for each ground-truth $g \in GT$ **do**
    Build an empty set for candidate anchors of the ground-truth g: $C_g \leftarrow \emptyset$;
    for each level $i \in [0, L-1]$ **do**
        Select $Topk$ anchors from A_i whose center points are closest to the center of ground-truth g based on L2 distance: S_i;
        $C_g = C_g \cup S_i$;
    end for
    Compute IoU between C_g and g: $F_g = IoU(C_g, g)$;
    Compute IoU threshold for ground-truth g: $Th_g = mean_\max(F_g)$;
    for each candidate anchor $c \in C_g$ **do**
        if $IoU(c, g) \geq Th_g$ and center point of c in g **then**
            $P = P \cup c$;
        end if
    end for
end for
$N = A - P$;
return P, N;

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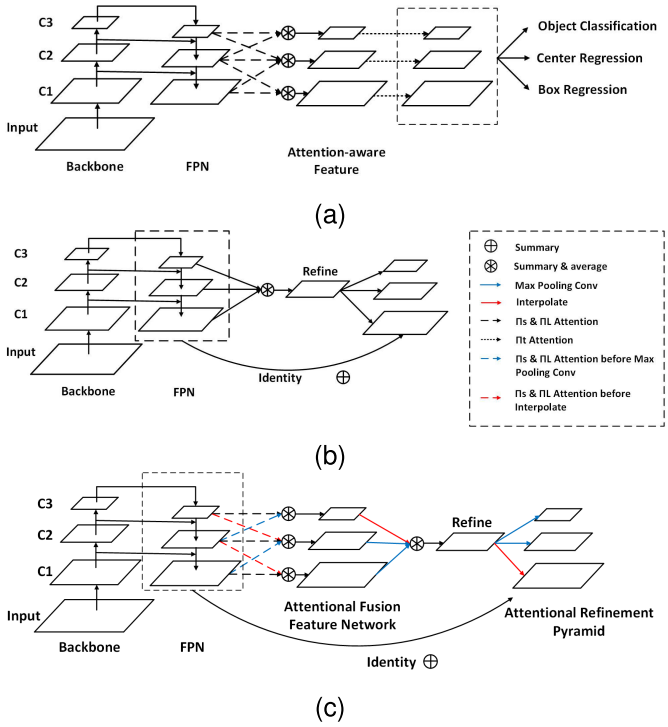


Fig. 6. Comparison of feature fusion structures. (a) DyHead employs scale, spatial, and task attention for feature attention calculation and fusion. (b) BFP uniformly scales feature maps of each layer to a specific size for fusion, refines them, and then restores them to the original size as attention feature maps overlaid onto the feature maps of the FPN structure. (c) BFPA first calculates the attention mechanism of adjacent layer feature maps, fuses them, scales them uniformly to a specific size, and, finally, restores them to the original size as attention feature maps overlaid onto the FPN structure's feature map.

The centralization score of the positive sample is calculated using the following formula:

$$e_i^* = \sqrt{\frac{\min(l^*, r^*)}{\max(l^*, r^*)} \times \frac{\min(t^*, b^*)}{\max(t^*, b^*)}}. \tag{2}$$

l^*, r^*, t^*, and b^* represent the distances from the center point of the anchor to the left, right, top, and bottom edges of the GT's horizontal bounding box. The normalized center distance is utilized to measure the distance between the anchor and its corresponding GT's center point. The value range is from 0 to 1. Here, it is easy to find that if the center point of the positive anchor is closer to the center point of the GT, our e_i^* value will be closer to 1. Therefore, the positive sample with higher e_i^* will be given a greater weight during training, which is more conducive to model learning and generating predictions close to the GT.

C. BFP With Attention

Due to the MMBA strategy generating only a square anchor box at each feature position, there is a higher requirement for generating feature maps for upstream tasks. Fig. 6 illustrates three distinct feature fusion networks, showing three feature layers for clarity, although the actual construction involves five layers. Dynamic head (DyHead) integrates spatial, scale, and task attention but often yields suboptimal results for detecting rotating object detection due to target scale variations and arbitrary orientations. The BFP augments the FPN feature map with global feature information but lacks detailed target characteristics. Therefore, we designed the BFPA module for the MMBA strategy, which adds attention mechanisms into spatial and shape information, increases information flow between adjacent layers, and fuses features from each layer to enhance the global semantic information of the feature map [as shown in Fig. 6(c)]. The attention mechanism enhances spatial and shape recognition, which is crucial for the positive learning of low-matching anchors introduced as positive samples in the MMBA strategy.

Assuming the given feature vector $F \in R^{S\times L\times C}$, we use two concatenated attention convolution operations called spatial attention mechanism and scale attention mechanism, denoted as Π_S and Π_L, operating on two scales, S and L, respectively.

The formula for the attention mechanism is

$$\mathrm{Fa}_i = \Pi_s(\Pi_L(F_i)). \tag{3}$$

1) Spatial Attention Π_S: We initially conduct spatial attention convolution on the feature map itself post FPN feature fusion, enhancing the spatial perception ability of the target through deformable convolution learning. This process comprises two stages: first, calculating the offset and mask through convolution; then, inputting the feature map and the obtained offset and mask into deformable convolution for learning

$$\Pi_s(F_i) = \sum_{k=1}^{K} \mathrm{DCN}(F_i;\, p_k + \Delta p_k;\, \Delta m_k) \tag{4}$$

where K represents the position of sparse sampling points and F_i represents the current layer feature map. Δp_k and Δm_k denote both the offset and mask learned by $\mathrm{Conv}_{3\times3}$, so $p_k + \Delta p_k$ represents the position after adding the offset calculated by the spatial self-attention mechanism and Δm_k represents an important value at the self-learning position. The value is calculated by the sigmoid function and ranges from 0 to 1. DCN indicates the use of deformable convolutions.

2) Scale Attention Π_L: We enhance scale perception by calculating scale-attention weights based on different importance semantics and then multiply them with the input feature map to obtain the output result

$$W_i = \sigma(f(\mathrm{Avg}P(H_i))) \quad \Pi_L = W_i \cdot H_i \tag{5}$$

where $\mathrm{Avg}P(\cdot)$ represents the calculation of average pooling, $F(\cdot)$ represents a $\mathrm{Conv}_{1\times1}$, and $\sigma(x) = \min(\max(((x + 3)/6), 0), 1)$ represents the hard-sigmoid function.

3) Attentional Feature Fusion Network: The AFFN calculates spatial attention and scale attention from the feature maps of adjacent layers. Then, it overlays them onto the current layer for averaging to obtain the feature fusion pyramid. Assuming that the feature maps of the feature fusion network (FFN) obtained after fusion are $\{C1, C2, C3, C4, C5\}$, the specific formula can be explained as follows:

$$C_i = \mathrm{avg}(\mathrm{Avg}P(\mathrm{Fa}_{i-1}) + \mathrm{Fa}_i + \mathrm{Up}S(\mathrm{Fa}_{i+1})).$$

Fa_{i-1} and Fa_{i+1} represent the feature maps of the lower and upper layers of Fa_i. $\mathrm{Avg}P$ and $\mathrm{Up}S$, respectively, resize the

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adjacent layers to the current Fa_i scale by the operation of average pooling and upsampling.

4) Attentional Refinement Pyramid: The refine level is a hyperparameter that selects a feature map's scale as the intermediate scale for the fusion of all layers in the AFFN (with $C3$ selected by default). All feature layers reshape to the size of the $C3$ feature map by max-pooling or upsampling. After that, the integrated feature map is obtained by adding all feature layers, which are then refined and convolved to produce a refined feature map:

$$C_{ig} = \frac{1}{L}(\mathrm{Avg}P(C_1) + \mathrm{Avg}P(C_2) + \mathrm{Avg}P(C_3) + \mathrm{Up}S(C4) + \mathrm{Up}S(C5)) \tag{6}$$

$$C_{rf} = \mathrm{RF}(C_{ig}). \tag{7}$$

L represents the total number of layers in the AFFN. The refinement operation, RF, utilizes a nonlocal block to capture long-distance information.

Finally, the refined feature map is adjusted back to its original scale using average-pooling and upsampling operations to obtain the ARP. By superimposing the ARP onto the FPN, we ensure that the convolutional layers in downstream tasks focus on the task-relevant parts of the feature data. Specific details are illustrated in Fig. 6(c).

D. Loss Function

The loss of AMMBA is a combination of multitask loss functions, including classification loss, localization loss, and center loss. The loss formula for each part is given as follows:

$$L_{\mathrm{cls}} = \frac{1}{N_{\mathrm{pos}}} \sum_{i=1}^{N} \mathrm{FL}\left(c_i^F, l_i^*\right) \tag{8}$$

$$L_{\mathrm{reg}} = \frac{1}{N_{\mathrm{pos}}} \sum_{i=1}^{N} e_i^* \cdot R\mathrm{IoU}\left(X_i^F, g_i^*\right) \tag{9}$$

$$L_{\mathrm{ct}} = \frac{1}{N_{\mathrm{pos}}} \sum_{i=1}^{N} \mathrm{BCE}\left(b_i^F, e_i^*\right) \tag{10}$$

where N_{pos} represents the number of positive samples, FL represents the focal loss for classification, c_i^F represents the predicted category, and l_i^* represents the category of GT (similar in L_{reg} and L_{ct}). RIoU represents the rotated IoU loss, and e_i^* is calculated as a weight to distinguish the quality of positive samples in 2. X_i^F and g_i^* represent the predicted and GT bounding boxes, respectively. The center loss BCE represents the binary cross-entropy loss.

The total loss calculation formula is given as follows:

$$L = \omega_1 L_{\mathrm{cls}} + \omega_2 L_{\mathrm{reg}} + \omega_3 L_{\mathrm{ct}}. \tag{11}$$

The default values for ω_1, ω_2, and ω_3 are 1, 2, and 1, respectively.

IV. Experiment

All experiments are conducted on publicly available RS image datasets, DOTA [40], HRSC [41], and DIOR-R [42], all of which utilize OBBs for labeling. The specifics of the experimental dataset, experimental parameter settings, experimental conditions, and experimental results will be elaborated on in this section.

A. Datasets

DOTA-v1.0 [40] encompasses 15 categories exhibiting significant differences in direction, shape, and size: plane (PL), baseball field (BD), bridge (BR), ground track field (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC). The image lengths range from 288 to 8115 pixels, and the widths range from 211 to 13 383 pixels. Due to the substantial variance in individual image sizes within the DOTA dataset, we uniformly cropped them to 1024 × 1024 with a gap of 200.

HRSC is a challenging aerial remote sensing image dataset introduced by Liu et al., focusing on ship categorization, including vesselsfrom ports and offshore areas, collected from Google Earth. The primary challenges in identification stem from the presence of numerous nearshore ships, characterized by a dense distribution, high overlap of annotation boxes, and diverse scale variations. The dataset comprises a total of 1061 images, distributed across training, validation, and test sets, with 436, 181, and 444 images, respectively. The image sizes range from 300 × 300 to 1500 × 900 pixels. To ensure consistency, we have rescaled them to 800 × 800 for both training and testing purposes.

DIOR-R [42], based on the DIOR [43] dataset, utilizes oriented bounding box annotation to address the objects. Comprising 23 463 optimal RS images and 190 288 target instances, the dataset features images with a size of 800 × 800 pixels and a spatial resolution ranging from 0.5 m to 30 m. This dataset is split into a training–validation set (11 725 images) and a testing set (11 738 images). DIOR-R serves as a large-scale benchmark dataset for oriented object detection in optical RS images.

B. Parameters and Experimental Details

All experiments are conducted based on the mmrotate1.x [59] framework, using PyTorch 1.8.1, MMCV 2.0.1, and MMEngine 0.10.3. By default, we build the baseline on rotated RetinaNet [3] with backbone ResNet50 [28] and FPN [32] structure. The model is trained and tested on a single RTX3090. The SGD optimizer is used with initial learning rate, momentum, and weight decay rates set to 0.01, 0.9, and 0.0001, respectively. Unless otherwise specified, DOTA-v1.0, HRSC, and DIOR-R are trained for 12, 36, and 12 epochs, respectively. For the MaxIoU strategy, three anchor box scales are used at each feature map position, with three different aspect ratios set for each size, totaling nine anchors. For ATSS [10], SASM [15], and the proposed MMA, MMBA, only one fixed square anchor box is generated at each position. Combined with the mentioned assignments, other anchor-based single-shot models also adhere to the default configuration. Except for the hyperparameter ablation experiment, the refine level of BFPA is set to 2, and the

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TABLE I

COMPARISON WITH THE STATE-OF-THE-ART DETECTOR BASED ON THE OBB TASK OF THE DOTA-V1.0. COLUMN "BB" MEANS THE BACKBONE. "AMMBA" DENOTES A COMBINATION OF MMBA STRATEGY WITH BFPA MODULE. "*" DENOTES MULTISCALE TRAINING AND TESTING WITH DATA AUGMENTATION, INCLUDING RANDOM FLIP AND RANDOM ROTATION. THE BEST AND SECOND-BEST RESULTS FOR EACH CATEGORY ARE HIGHLIGHTED IN BOLD FONT

Methods	BB	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP(%)
FCOS-CSL [44]	R-50	88.24	74.87	41.27	61.03	79.52	78.35	87.19	90.88	81.50	84.53	54.70	62.65	62.84	68.45	46.50	70.83
RetinaNet-PSC [45]	R-50	89.32	82.29	37.92	71.52	78.40	66.33	78.01	90.89	84.21	80.63	60.22	64.73	59.69	68.37	53.85	71.09
P-RSDet [46]	R-101	88.58	77.84	50.44	69.29	71.10	75.79	78.66	90.88	80.10	81.71	57.92	63.06	66.30	69.77	63.13	72.30
FCOS-KLD [47]	R-50	89.05	73.73	49.17	57.86	79.54	77.91	87.41	90.87	83.23	82.42	58.97	61.83	62.91	72.69	59.44	72.47
FCOS-PSCD [45]	R-50	89.06	73.61	49.03	62.34	75.18	77.69	88.00	90.85	82.63	72.97	61.48	64.20	65.77	72.88	67.88	72.90
R^3Det* [48]	R-152	89.24	80.81	51.11	65.62	70.67	76.03	78.32	90.83	84.89	84.42	65.10	57.18	68.10	68.98	60.88	72.81
DRN* [49]	H-104	**89.71**	82.34	47.22	64.10	76.22	74.43	85.84	90.57	86.18	84.89	57.65	61.93	69.30	69.63	58.48	73.23
SRDF [50]	R-101	87.55	84.12	52.33	63.46	78.21	77.02	88.13	90.88	86.68	85.58	47.55	64.88	65.17	71.42	59.51	73.50
PARDet [51]	R-50	87.97	77.67	49.65	67.16	79.06	75.65	87.21	90.90	84.76	83.06	61.61	66.94	65.75	66.93	58.21	73.50
Gliding Vertex [52]	R-101	89.64	**85.00**	52.26	**77.34**	73.01	73.14	86.82	90.74	79.02	86.81	59.55	**70.91**	72.94	70.86	57.32	75.02
Oriented RepPoints [23]	R-50	88.77	80.47	54.54	72.64	80.39	80.18	87.74	**90.90**	84.21	83.86	63.97	68.55	70.54	70.62	51.55	75.26
BBAVectors* [53]	R-101	88.63	84.06	52.13	69.56	78.26	80.40	88.06	90.87	87.23	86.39	56.11	65.62	67.10	72.08	63.96	75.36
Oriented RCNN [18]	R-50	89.46	82.12	54.78	70.86	78.93	83.00	88.20	**90.90**	**87.50**	84.68	63.97	67.69	74.94	68.84	52.28	75.87
SODC [54]	R-50	88.54	**84.72**	51.21	70.18	79.43	80.82	88.56	90.66	**87.28**	86.49	55.37	64.59	68.22	71.03	64.85	76.06
DAL* [17]	R-50	89.69	83.11	55.03	71.00	78.30	81.90	88.46	90.89	84.97	87.46	64.41	65.65	76.86	72.09	64.35	76.95
R^3DET-DCL* [19]	R-152	89.26	83.60	53.54	72.76	79.04	82.56	87.31	90.67	86.59	86.98	67.49	66.88	73.29	70.56	**69.99**	77.37
F-ABFL w AST* [55]	R-152	**89.73**	83.85	56.56	70.78	**81.85**	84.61	**88.71**	90.85	85.74	86.50	**67.91**	**71.28**	**78.61**	**80.69**	65.56	78.88
RetinaNet	R-50	89.41	76.81	40.88	67.54	77.50	62.65	77.54	90.89	82.31	81.96	58.15	61.56	56.47	63.71	38.97	68.42
RetinaNet+AMMBA	R-50	89.30	77.05	50.26	71.23	80.56	79.40	87.56	**90.90**	84.16	85.22	64.34	66.54	66.44	69.38	52.58	74.33
RetinaNet+AMMBA*	RX-101	89.14	82.53	**57.80**	77.10	**82.67**	**85.72**	88.53	90.77	87.20	**87.82**	64.74	69.39	77.72	77.56	69.96	**79.24**
S^2A-Net [7]	R-50	89.11	82.84	48.37	71.11	78.11	78.3	87.25	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94	74.12
S^2A-Net+AMMBA	R-50	89.29	81.50	49.68	72.70	79.60	78.40	86.16	**90.90**	83.83	85.12	63.85	66.60	66.93	69.32	56.37	74.68
S^2A-Net+AMMBA*	Swin-L	89.62	84.11	**61.04**	**81.70**	79.85	**85.45**	**88.96**	90.84	86.97	**87.73**	**73.07**	68.92	**77.86**	**79.15**	**78.44**	**80.91**

*SODC is trained by 40 epochs.
*R-50, R-101, R-152, RX-101, H-101, Swin-L denote ResNet-50 [28], ResNet-101 [28], ResNet-152 [28], ResNeXt-101 [30], Hourglass-101 [56] and Swin-Transformer-Large [31].
*All methods use the FPN structure.

topk of MMBA is set to 9. All models are trained with random flipping for data augmentation, and multiscale training procedures additionally incorporate random rotation.

C. Comparison With State-of-the-Art Detectors

1) Results on DOTA: Table I presents a comparative analysis between our proposed module and other state-of-the-art models on the DOTA dataset. Our module achieves an accuracy of 74.68% mAP when based on RetinaNet, utilizing ResNet50 as the backbone, and operating at a single scale. Remarkably, this result outperforms single-shot detectors with multiple cascade detection heads, such as S^2A-Net and R^3Det, even without any additional data augmentation techniques. With the incorporation of data augmentation and multiscale training, our method achieved an mAP accuracy of 80.91%, surpassing single-shot detectors and comparable to some two-stage detectors. It is obvious that our method outperforms other methods on categories with arbitrary angles, such as BR, LV, HA, and HC.

This improvement is attributed to MMBA's adoption of more positive samples with good location and object-enclosure potential but lower IoUs with GTs, which were usually considered negative samples in previous approaches. This enables the model to be trained more efficiently leveraging more positive samples. With the assistance of centerness scores from the BA strategy, it is easier for the model to distinguish between better and worse training samples, leading to more stable and effective training. Notably, the adoption of the proposed AMMBA led to significant improvements in several categories with initially poor accuracy, such as SBF and GTF. This confirms the effectiveness and advancement of our AMMBA in this article.

2) Results on HRSC: The ship objects in HRSC exhibit arbitrary angles, prompting previous methods to frequently employ preset multishape and multiratio anchors to enhance the recall rate of positive samples. According to the findings outlined in Table II, our AMMBA approach achieves optimal accuracy, even with a reduced number of anchors employed in the RetinaNet framework. Without bells and whistles, when integrated with the S^2A-Net framework, our method achieves an accuracy of 90.50%. This accomplishment highlights the innovative and efficient nature of the BFPA and MMBA strategy, showcasing that our approach can achieve state-of-the-art results without resorting to unnecessary hyperparameter adjustment experiments.

TABLE II

COMPARISON WITH THE STATE-OF-THE-ART DETECTOR ON HRSC

Models	Backbone	Epochs	mAP(%)
RetinaNet [3]	R-101	6×	86.40
RRD [57]	VGG16	3×	84.30
Roi-Trans. [58]	R-101	3×	86.20
Gliding Vertex [52]	R-101	3×	88.20
R^3Det [48]	R-101	3×	89.26
DAL [17]	R-101	3×	89.77
reppoint [4]	R-101	3×	85.16
AOPG [42]	R-101	3×	90.34
S^2A-Net	R-101	3×	90.17
S^2A-Net+AMMBA(Ours)	R-101	3×	**90.50**

TABLE III

COMPARISON WITH THE STATE-OF-THE-ART DETECTOR ON DIOR-R. USE THE RESNET50-FPN AS THE BACKBONE NETWORK ARCHITECTURE FOR ALL EXPERIMENTS

Models	mAP(%)	Models	mAP(%)
RetinaNet [3]	57.6	FR-O [1]	59.5
Gliding Vertex [52]	60.1	Roi-Trans [58]	63.9
Oriented Reppoint [23]	63.0	PARDet [51]	63.4
SASM-BA [15]	62.9	S^2A-Net [7]	61.4
RetinaNet+AMMBA(Ours)	63.8	S^2ANet+AMMBA(Ours)	**64.6**

3) Results on DIOR-R: The DIOR-R dataset comprises 20 categories, with RS targets exhibiting a wide range of arbitrary directions and scales. This variability presents significant challenges for the model's regression task learning. Among

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TABLE IV
COMPARISON WITH ASSIGNMENT STRATEGIES THAT INVOLVE IOU THRESHOLD ADJUSTMENT ON THE DOTA DATASET. ALL EXPERIMENTS ARE CONDUCTED USING THE RETINANET-RESNET50-FPN NETWORK ARCHITECTURE

Models	FPS	GFLOPS	Params(M)	mAP(%)
RetinaNet(baseline)	56.9	215.92	36.42	68.42
ATSS-BA	**57.2**	**207.96**	**36.05**	73.37
SASM-BA	55.0	**207.96**	**36.05**	72.37
AMMBA(Ours)	56.1	209.62	38.23	**74.33**

recent similar methods, as listed in Table III, S^2A-Net with our AMMBA achieved the highest accuracy at 64.6%, which demonstrates superior performance of the proposed MMBA and BFPA methods.

D. Comparison With Other Assignment Strategies

1) Comparison With the Assignment Strategies Involving IoU Threshold Adjustment: To further demonstrate the superiority of our proposed method, we compare it with ATSS and SASM across four aspects: model speed, computational complexity, number of parameters, and accuracy. As shown in Table IV, our method achieves the best detection results, with only a slight tradeoff in reasoning speed and a modest increase in the number of parameters. The reasons perhaps are given as follows:

1) compared to the mean and variance calculations used by ATSS and SASM, the mean-max function involves more complex operations, resulting in additional computational overhead and
2) the difference in the number of matched positive samples slightly affects inference speed. However, these minor compromises in speed and memory usage are acceptable given the significant improvements in detection accuracy.

2) Comparison With Other Assignment Strategies: Experimental results of HRSC are shown in Table V. Due to the large aspect ratio of HRSC vessels, which poses a major challenge to angle classification learning, CSL [44] performs worse than the baseline, with an accuracy drop of 1.3%. Compared with the baseline, the accuracies of ATSS, DAL, SASM, and our MMA are significantly improved, which confirms the feasibility of dynamic threshold design based on statistical methods. Among these statistical methods, our MMA achieves the best improvement (89.5%), which means that our MMA is better than other dynamic threshold algorithms. Meanwhile, with the proposed balancing assignment strategy, our MMBA (MMA with BA) can further improve its learning ability for positive samples, especially high-quality positive samples, thereby further improving the recognition accuracy of the detector.

E. Inference Speed Versus Accuracy

A comparison of inference speed and accuracy was carried out on DOTA-v1.0 using ResNet50-FPN as the feature extraction network, with a batch size set to 1, as depicted in Table VI. For consistency, we conducted all experiments on an RTX 3090. It is evident that our method achieves optimal performance among single-stage models, with only a slight decrease in speed. While its accuracy remains comparable to that of two-stage models, its inference speed surpasses that of those models. The result demonstrates that AMMBA achieves an effective balance between speed and accuracy.

TABLE V
VALIDATE THE EFFECTIVENESS OF THE MMA STRATEGY ON THE HRSC DATASET. "6×" DENOTES TRAINING FOR 72 EPOCHS, ALL UTILIZING THE RETINANET FRAMEWORK WITH RESNET50 AS THE BACKBONE

Models	Backbone	Epochs	mAP(%)
RetinaNet(baseline)	R-50	6×	84.8
CSL	R-50	6×	83.5
ATSS	R-50	6×	86.1
DAL	R-50	6×	88.6
SASM	R-50	6×	88.8
baseline+MMA	R-50	6×	89.5
baseline+MMBA	R-50	6×	**90.2**

TABLE VI
COMPARISON OF INFERENCE SPEED AND ACCURACY WITH STATE-OF-THE-ART DETECTORS ON THE DOTA DATASET. USE RETINANET-AMMBA WITH RESNET50-FPN BACKBONE NETWORK ARCHITECTURE FOR ALL EXPERIMENTS

Models	Framework	FPS	mAP(%)
RetinaNet	One-stage	56.9	68.42
S^2A-Net	One-stage	55.1	74.12
Oriented RCNN	Two-stage	53.8	75.87
Roi-Trans	Two-stage	44.4	74.61
RetinaNet+AMMBA(Ours)	One-stage	56.1	74.33

V. ABLATION ANALYSIS AND DISCUSSION

To validate the effectiveness of the proposed module and streamline factors, we start with Rotated RetinaNet as the baseline to test the MMBA strategy and the BFPA module.

A. Ablation Experiments of Hyperparameters

1) Hyperparameter Topk of MMBA: The hyperparameter *topk* is the unique hyperparameter in MMBA, determining the number of candidate anchor boxes selected for each GT on a single feature layer. To assess its impact on experimental results, we explore various parameter values, specifically $topk = [5, 7, 9, 11, 13]$. The experimental outcomes are detailed in Table VII. Notably, when $topk = 9$, it achieves the best mAP. Increasing the value includes more low-quality anchors, leading to erroneous labeling of poorly matched anchor boxes as positive samples, thereby causing learning instability. Conversely, when it is smaller, accuracy declines due to fewer anchor boxes being designated as positive samples, resulting in inadequate sample learning. In this ablation experiment, the maximum difference in accuracy among the experimental results for different *topk* values does not exceed 0.63, indicating that the experiment is not sensitive to hyperparameter fluctuations within the selected range.

2) Hyperparameter Refine Level of the BFPA: The refine level specifies the layer of the feature map size that the readjusted feature map should match, also called the intermediate size. In this experiment, the MaxIoU assignment strategy

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TABLE VII

ANALYZING THE INFLUENCE OF VARIOUS *topk* VALUES IN MMBA ON THE DOTA DATASET, UTILIZING RETINANET WITH RESNET50 AND FPN AS THE FRAMEWORK

Topk	5	7	9	11	13
mAP(%)	73.15	73.38	**73.74**	73.11	73.44

TABLE VIII

ABLATION EXPERIMENTS ON THE REFINE LEVEL HYPERPARAMETERS IN THE BFPA MODULE WERE CONDUCTED USING THE DOTA DATASET, EMPLOYING THE RETINANET MODEL CONFIGURED WITH RESNET50

refine level	neck	mAP(%)
baseline(None)	FPN	68.42
0	FPN+BFPA	68.99
1		69.35
2		**69.64**
3		69.46
4		68.76

TABLE IX

ABLATION EXPERIMENTS OF THE JOINT MMBA STRATEGY AND BFPA MODULE OF AMMBA ON THE DOTA DATASET, EMPLOYING THE RETINANET FRAMEWORK CONFIGURED WITH RESNET50 AS BASELINE. THE IMPROVED ACCURACY IS CALCULATED BASED ON THE BASELINE. MMBA INCLUDES THE USE OF THE CENTERNESS BRANCH, WHILE MMA DOES NOT

	MaxIoU	MMA	MMBA	Dyhead	BFP	BFPA	mAP(%)
1	✓	-	-	-	-	-	68.42 **(+0)**
2	-	✓	-	-	-	-	71.48 **(+3.06)**
3	-	-	✓	-	-	-	73.74 **(+5.32)**
4	✓	-	-	✓	-	-	61.89 **(-6.53)**
5	✓	-	-	-	✓	-	69.13 **(+0.71)**
6	✓	-	-	-	-	✓	69.46 **(+1.04)**
7	-	✓	-	✓	-	-	65.55 **(-2.87)**
8	-	✓	-	-	✓	-	71.41 **(+2.99)**
9	-	✓	-	-	-	✓	71.93 **(+3.51)**
10	-	-	✓	✓	-	-	73.14 **(+4.72)**
11	-	-	✓	-	✓	-	73.29 **(+4.87)**
12	-	-	✓	-	-	✓	74.33 **(+5.91)**

is adopted. We designed five experiments, selecting different layer sizes as the refine level for experimentation. The results are presented in Table VIII.

It is easily observed that regardless of the chosen layer size for feature fusion scaling, the accuracies compared with the baseline model all improve. This demonstrates the effectiveness of the BFPA module. When the refine level is set too high or too low, it implies that there is a significant gap between each layer and the intermediate layer size. This excessive difference causes the feature fusion module to overly emphasize either semantic or detail information, resulting in minor fluctuations in recognition accuracy. However, the maximum and minimum experimental accuracies differ by no more than 0.7%, indicating that this hyperparameter is robust, rendering the module essentially free of hyperparameter tuning.

B. Ablation Experiments of MMBA and BFPA

1) MMBA Strategy: The experimental results in Table IX, shown in lines 1–3, are obtained without employing any feature fusion techniques except for FPN. Here, our MMA strategy without centerness branches assigns the same weight to all positive samples for calculating losses, and the proposed MMBA with centerness branches adopts dynamic weight assignment to enhance high-quality positive sample learning (described in Section III-B). When the MaxIoU strategy is replaced with the MMA strategy, a notable increase in accuracy is observed, rising from 68.42% to 71.48%. This improvement amounts to a total enhancement of 3.06% (lines 1 and 2), which means that our MMA is more effective in distinguishing positive and negative samples to promote model training.

For MMBA, the mAP achieves an impressive 73.74%. This represents a 2.26% increase compared to MMA (lines 2 and 3), significantly improving the accuracy of single-stage models, which means that the dynamic weights from the centerness branch can balance the impact of positive samples and then further enhance high-quality positive sample learning balancedly. The above two comparisons show that in our MMBA, the new sample assignment strategy (*i.e.,* MMA) and the dynamic weight of positive samples are indeed effective for the object detection of RS.

2) BFPA Module: The experimental results are shown in lines 4–6 of Table IX. To fairly assess the effectiveness of the BFPA module and eliminate the influence of MMBA on accuracy, we initially utilized the MaxIoU strategy for baseline experiments. When only using the DyHead [36] module, the accuracy is 6.53% lower than the baseline, indicating its unsuitability for direct transplantation into the oriented object detection due to the additional angle parameter in bounding box regression. Alternatively, solely employing the BFP [35] module results in a 0.71% improvement over the benchmark. In contrast, when integrating the BFPA module, accuracy significantly improves from 68.42% to 69.46% compared to the baseline model. This demonstrates that even when the BFPA module is added separately to different models without MMBA, it significantly improves the ability to extract target features by enhancing global and local attention information.

3) Joint Effect of MMBA and the BFPA Module in AMMBA Strategy: The experiments involving MMA and feature fusion networks are shown in lines 7–9. Even when combined with MMA, DyHead [36] underperforms compared to the baseline. Using BFP [35] improves accuracy by 2.99%. In contrast, the BFPA module proposed in this study achieves the highest improvement at 3.51% (line 9), providing more precise target information for downstream sample partitioning through global and local attention calculations. The experiments of the joint effect of MMBA and the BFPA module are recorded in lines 10–12. When our MMBA combines with the DyHead and BFP modules (lines 10 and 11), accuracy improves compared to the baseline but does not surpass the result without any feature fusion module (line 3). This indicates that only providing global or spatial and shape attention mechanisms can reduce accuracy by introducing inappropriate feature information. However, combining the BFPA module with the MMBA strategy surpasses the baseline by 5.91% and improves by 0.59% compared to line 3. The BFPA module focuses on global and spatial shape attention mechanisms, effectively capturing global semantic information and providing more

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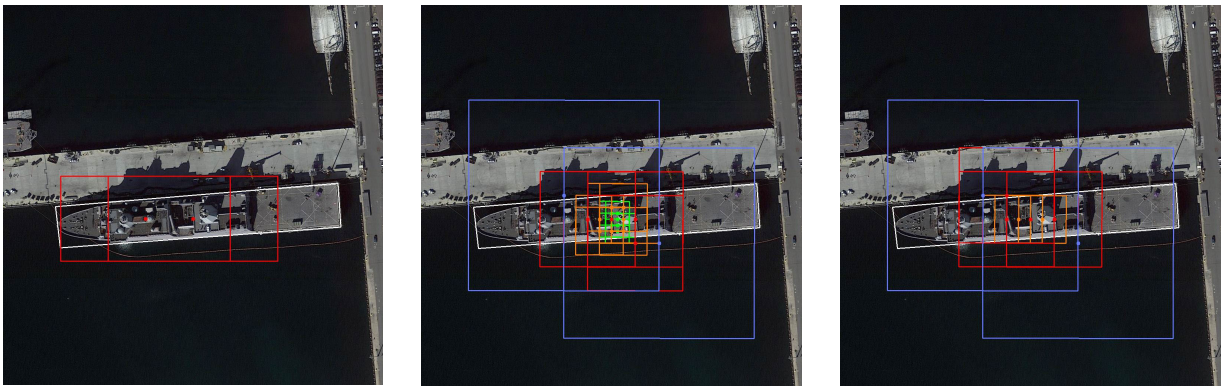


Fig. 7. Visualization of anchors selected as positive samples using the MaxIoU, SASM, and proposed MMA methods. The purple, red, orange, and green anchors are selected at the fifth, fourth, third, and second feature layers, respectively. (No anchors are selected at the first layer due to the relatively large-scale ship in the image. In addition, due to accuracy errors in the encoding and decoding of anchor boxes, some anchor box center points did not align with the GT during the drawing process.) (a) MaxIoU. (b) SASM. (c) Ours.

TABLE X

EFFECTIVENESS OF THE CENTERNESS BRANCH USED IN THE BA PROCESS IS VALIDATED ACROSS DIFFERENT CENTER DISTANCE-BASED POSITIVE AND NEGATIVE SAMPLE ASSIGNMENT STRATEGIES ON THE DOTA DATASET

Method	SV	LV	SH	HE	mAP(%)
ATSS	76.48	71.35	82.61	39.85	70.64
ATSS-BA	79.90	79.26	87.60	49.71	73.37 **(+2.73)**
SASM	76.76	71.59	82.73	38.1	70.26
SASM-BA	79.88	78.61	87.38	43.98	72.37 **(+2.11)**
MMA	76.95	71.40	81.67	45.55	71.48
MMBA	80.16	79.30	87.33	50.64	73.74 **(+2.26)**

TABLE XI

UNIVERSALITY STUDY OF THE TWO MODULES PROPOSED IN THIS ARTICLE ON OTHER ANCHOR-BASED SINGLE-SHOT ROTATED OBJECT DETECTORS BASED ON THE DOTA-V1.0 DATASET

Models	Feature Fusion	allocate strategy	mAP(%)
RetinaNet-KLD	FPN	MaxIoU	70.34
	FPN+BFPA	MMBA	71.65 **(+1.31)**
R^3Det	FPN	MaxIoU	69.80
	FPN+BFPA	MMBA	71.27 **(+1.47)**
S^2A-Net	FPN	MaxIoU	73.20
	FPN+BFPA	MMBA	74.68 **(+1.48)**

accurate target features for MMBA. What is more, the centerness branch in MMBA also benefits from spatial attention in BFPA. The proposed AMMBA strategy proposed in this study enables RetinaNet to achieve optimal accuracy on the single-scale DOTA dataset.

4) Analyze the Impact of the BA: Based on Table X, the BA strategy, which employs a centerness branch, plays a significant role in the regression label assignment methods. To further investigate, we conduct comparative experiments on the accuracy results with and without the centerness branch of the BA strategy in the three allocation methods of ATSS, SASM, and MMA. These methods rely on center distance to build candidate anchors' sets and ensure that the centers of positive samples remain within the GT boxes. The MaxIoU strategy does not have this restriction, so it is not discussed in this experiment. When IoU values are similar, anchors with shorter center distances tend to perform better in localization. The BA leverage increasing the weight of the anchors during loss calculation helps the model distinguish between high-quality positive samples and suboptimal ones, thereby suppressing the gradient of samples with poor overlap. In Table X, we select several object categories with significant recognition improvement, demonstrating that the addition of the centerness branch alone enhances experimental accuracy across different label assignment strategies. With the combination of BA procedure, MMBA improves accuracy by 2.26% compared to MMA. This result was achieved without utilizing the BFPA module, yet it still demonstrates excellent performance in single-stage approaches. It indicates that the BA-based learning of the relationship between anchors and GT boxes in center localization markedly improves the model's regression performance for high-quality positive samples.

C. Plug-and-Play Capability of the Module

This article combines the two proposed modules with mainstream anchor-based single-shot rotation object detectors, namely, S^2A-Net [7], Retinanet-KLD [47], and R^3Det [48] models, for experiments on the DOTA-v1.0 dataset. The results are presented in Table XI. It is evident that when the single-shot model integrates BFPA and MMBA strategy, there is a significant improvement in mAP by at least 1.3%. This demonstrates that the two proposed modules (MMBA and BFPA) can be seamlessly applied to the anchor-based single-shot detectors, demonstrating their versatility and plug-and-play capabilities.

D. Visualization Comparison and Analysis

To clearly demonstrate the differences between the proposed method and other sample assignment strategies, we select a ship sample (outlined by the white box) with a large aspect ratio and apply different methods to select positive samples for comparison, as shown in Fig. 7. In the case of Fig. 7(a), the MaxIoU method selects only two potential anchors as positive samples, leading to insufficient training on positive

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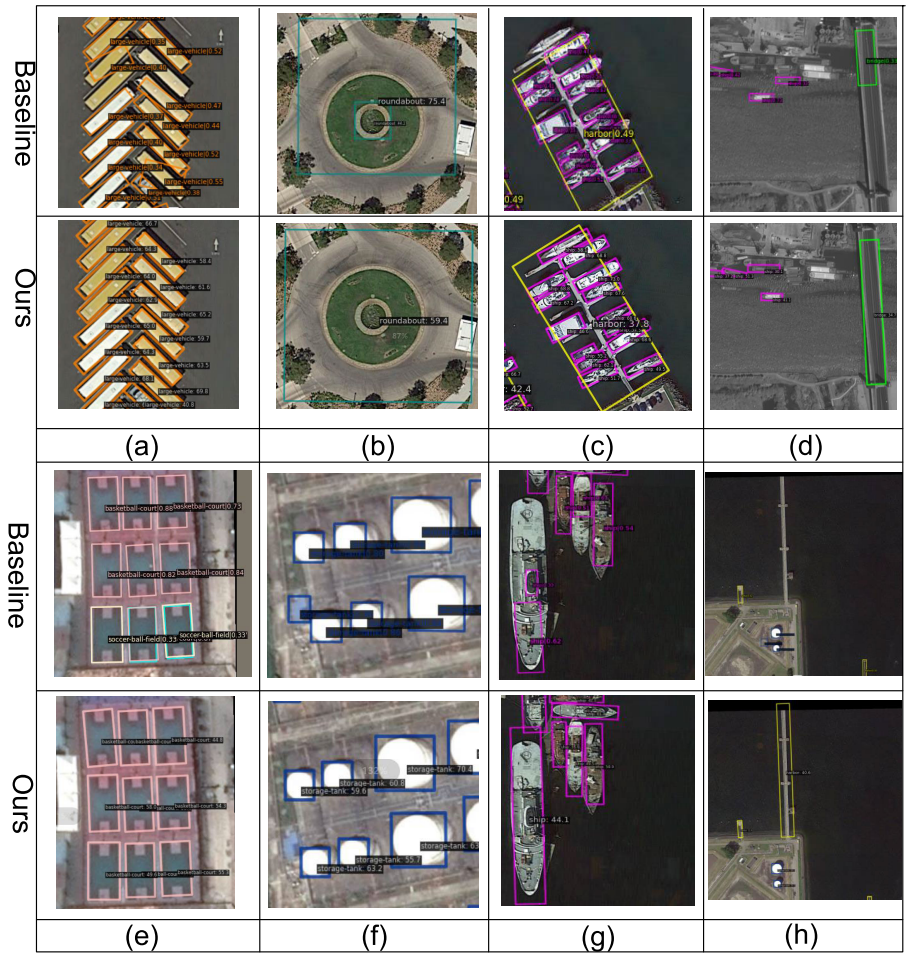


Fig. 8. Visualization results on the DOTA-v1.0 test set. Compare with Retinanet (Baseline), the better detection results of our AMMBA for different categories, including: (a) large vehical, (b) roundabouts, (c) harbors and ships, (d) bridges, (e) basketball courts, (f) storage tanks, (g) ships, and (h) harbors. Notably, our AMMBA model demonstrates superior predictive performance.

samples and poor performance in ship classification. In the case of Fig. 7(b), the SASM method selects 24 potential anchors and introduces narrow receptive field anchors that cannot adequately cover the ship (e.g., green anchors), which increases computational cost and negatively affects training stability. In the case of Fig. 7(c), our method strikes a balance between sufficient anchors and reduced redundant anchors. After applying the mean-max function, only ten anchors are retained as positive samples.

The final inference results demonstrate more robust detection performance after training. Fig. 8 displays the visualizations on the DOTA dataset. Compared with RetinaNet as the baseline, our proposed method demonstrates superior localization accuracy in visualization, as shown in Fig. 8(a) and (c). What is more, utilizing a single anchor box at each position helps prevent duplicate predictions, as shown in Fig. 8(b). More importantly, the feature maps of each layer are enriched with global semantic information after passing through the BFPA module, reducing erroneous predictions caused by solely focusing on local features shown in Fig. 8(d), (g), and (h). This results in a significant reduction in the occurrence of ambiguous prediction outcomes and false detection rates, as shown in Fig. 8(e) and (f), respectively. This visualization further verifies the effectiveness of our approach in object detection in RS images. Some visualization results about HRSC and DIOR-R datasets are presented in Figs. 9 and 10, respectively.

E. Discussion

This article proposes an AMMBA strategy. Extensive experiments on three classic optical RS datasets demonstrate that our method can be applied to target detection tasks across any optical RS dataset. However, our method still has limitations. As mentioned in Section IV-D, considering that the mean-max function involves complex operations, the AMMBA model's detection speed is slightly affected. In the next stage, we plan to explore faster dynamic IoU threshold generation methods to improve the detection accuracy in a faster fashion. In addition, for the RS dataset DOTA, this article obtains the optimal *topk* value of the proposed MMBA via lots of iterative exploration experiments. Therefore, exploring automatic generation methods for obtaining the optimal *topk* value for different datasets will also become the next study. Future work will focus on addressing these limitations by designing more detailed and professional exploration experiments while exploring a wider range of application scenarios.

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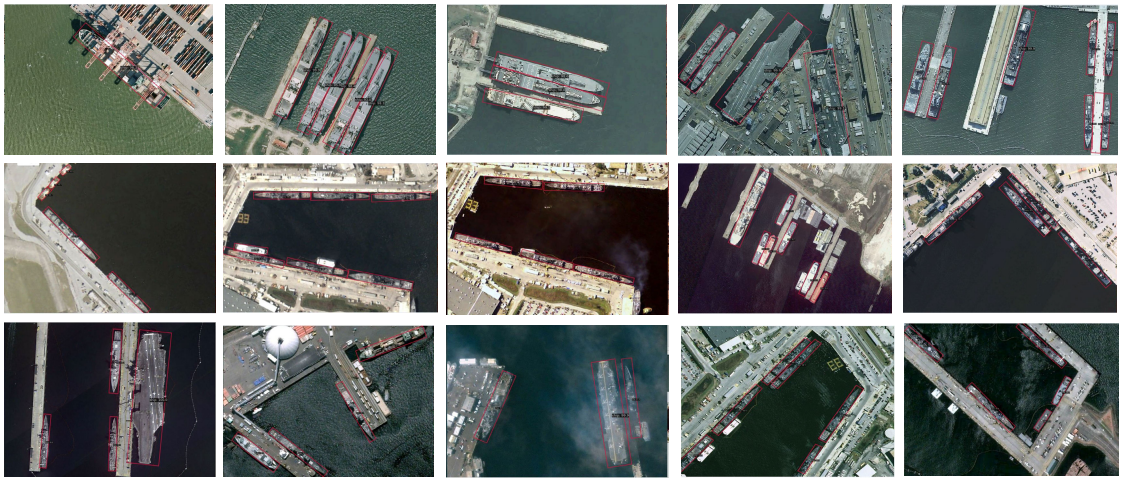


Fig. 9. Visualization results on the HRSC dataset using S2ANet-AMMBA with ResNet101-FPN backbone.



Fig. 10. Visualization results on the DIOR-R dataset using S2ANet-AMMBA with ResNet101-FPN backbone.

VI. Summary

In this article, we propose a simple and effective AMMBA strategy to address the limitations of existing fixed and soft-label assignments, improving the learning quality of positive samples during training. AMMBA combines MMBA and the BFPA. Among MMBA, the MMA ensures that each GT is matched with enough positive samples, enhancing model stability, while the BA further helps the detector focus on high-quality positive samples during training. Meanwhile, BFPA, by utilizing spatial and global attention, provides a more accurate feature representation. Ablation experiments confirm the effectiveness of MMBA and BFPA, while comparisons with other strategies demonstrate the superiority and versatility of AMMBA across various RS datasets, achieving top performance on DOTA-v1.0, HRSC, and DIOR-R. The method proposed in this article can be easily plugged into a one-stage anchor-based object detector, enhancing its performance.

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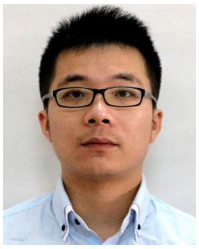
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